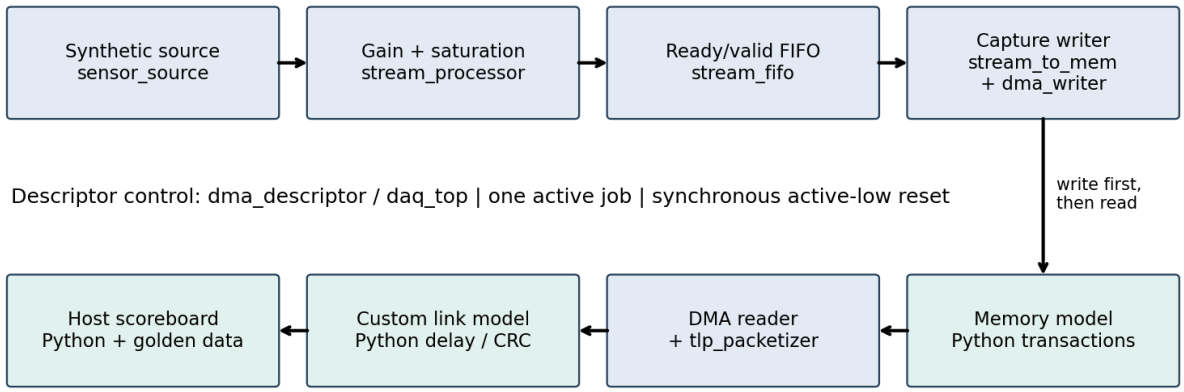
Verified RTL simulation

Publication recommendation: ready as a scoped simulation portfolio

63 tests PASS; all 33 E2E scenarios MATCH. A fresh no-cache Docker tool build and source-only workspace reproduce lint, the complete test suite and generic synthesis. No new production RTL or model features were added during this review.

Verified DAQ simulation architecture



Blue: synthesizable SystemVerilog. Green: simulation-only Python.
No DDR PHY, real PCIe endpoint, device timing or concurrent continuous capture.

Verification	Executed result
Production tests	39 pytest + 24 cocotb; 0 failures/skips. 33 matrix rows are inside these tests.
Independent challenges	Independent integer/CRC fixtures; six isolated mutations detected. An assertion-aborted mutant uses a bounded watchdog.
Toolchain	Verilator 5.040; cocotb 2.0.0; pytest 8.4.2; Yosys 0.33. Actual versions/return codes retained.
Generic synthesis	No latches/combination loops; 1,447 generic flip-flops; 42-cell longest topological combinational path.

Device timing closure remains BLOCKED. There is no selected FPGA/speed grade, placed-and-routed design, vendor timing model or setup/hold slack. The illustrative 100 MHz constraint is not a result. DDR is behavioral; the link uses a custom envelope/CRC, not a real PCIe endpoint.

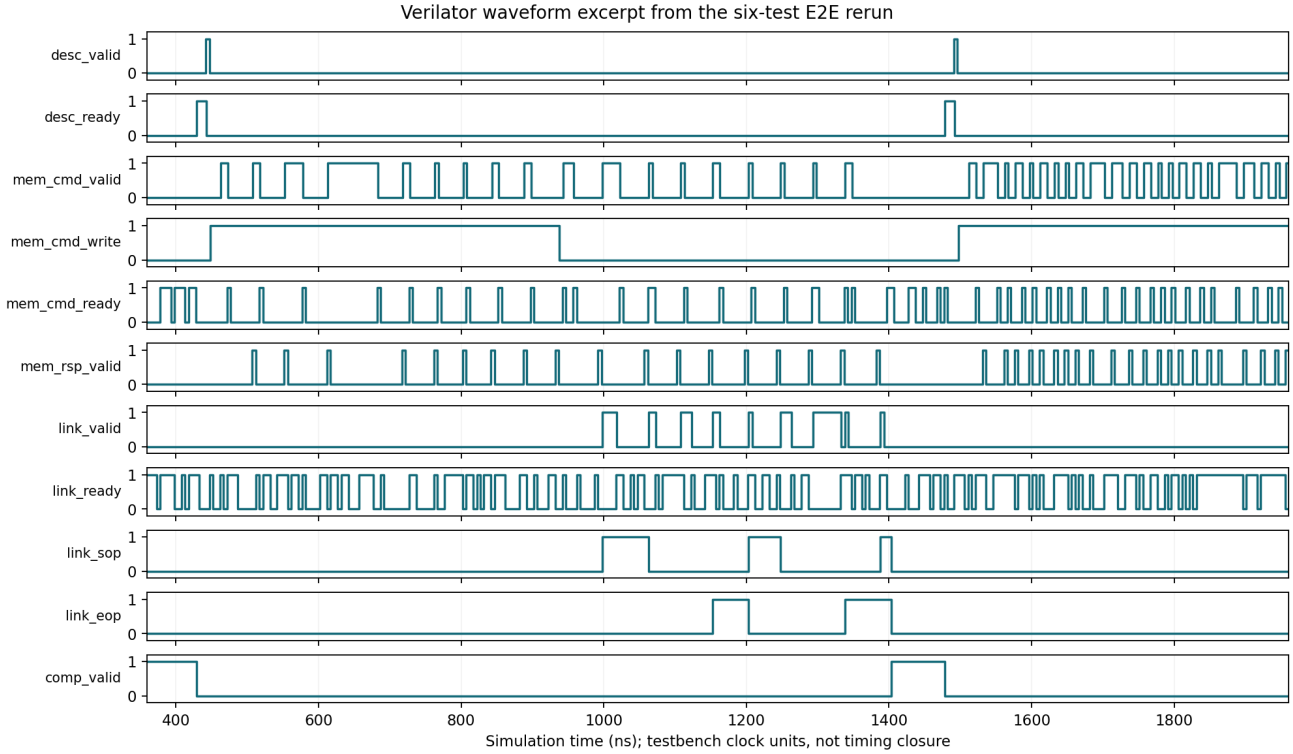
Source and command evidence: docs/VERIFICATION_REVIEW.md; docs/SYSTEM_AUDIT.md; results/release-audit/; synthesis/reports/. No GitHub push or remote CI run occurred.

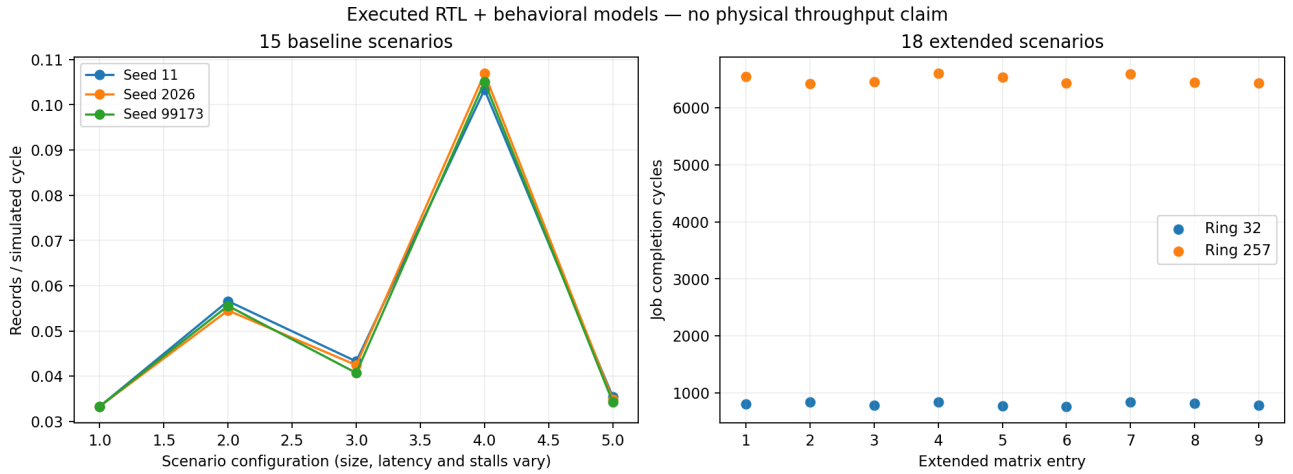
Debugging and real traces

Reproduction defects fixed; functional requirements unchanged

Prior verified fixes: 4 KiB packet crossing at 0x1ff8; rejection of the final legal 8-byte record; host-context loss when completion preceded delayed delivery; and a runner that returned success despite failed cocotb XML.

Independent-review defect: clean builds lacked results/system-audit/, causing extended tests to fail. The runner and writer now initialize it. A second clean source copy with a new tool image passes. Fault injection does not modify production sources.





Plots are generated from actual VCD transitions and recorded benchmark JSON. Axes represent simulated time/cycles, not physical timing or bandwidth. Full VCD and SVG/PNG provenance: results/waveforms/ and portfolio/assets/.

Separate PCB: remaining FAILs

Rev 0.3.1 Pi carrier is not the FPGA DAQ board

Requirement	Result / unchanged acceptance
Voltage budget	3.130447-3.487610 V versus 3.135-3.465 V; margins -4.553 / -22.610 mV. FAIL.
Thermal minimum copper	6/24 cases fail at 30 um outer copper. Fine worst 160.303 C versus 125 C; margin -35.303 C.
Current impedance	Original broad high corner 106.126 ohm versus 99 ohm; local wide-gap fanout 104.135 ohm. FAIL.
Native CAD	Clean rebuild: ERC/DRC/unconnected/parity all zero. Geometric success does not qualify electrical or manufacturing gates.

Voltage roots: low feedback/divider/leakage plus copper, ground/contact and dynamic reserves; high PSM feedback plus positive dynamic allowance. A divider-only solution cannot satisfy both bounds. PWM high margin is +10.847 mV; PSM is the failing upper case. The 16 uF stress is outside the datasheet PSM row's explicit 22 uF condition.

Thermal correction: 35 um nominal copper was inconsistent with the 30 um minimum used for DC. All 24 cases were rerun without lowering losses or ambient. Original 159.381 C becomes 157.389 C with finer grid at 35 um, but 160.303 C at 30 um. The sixth failure remains 125.303 C on the fine grid.

At the worst original hotspot, regulator board rise is 52.567 C; SSD coupling 22.379 C; inductor 8.536 C; copper 6.753 C; simplified package path 7.459 C. Weak convection and local heat dominate. Exact uniform-source and superposition checks pass. Above absolute maximum / near shutdown, the linear extrapolation is a rejection indicator, not sustainable operation.

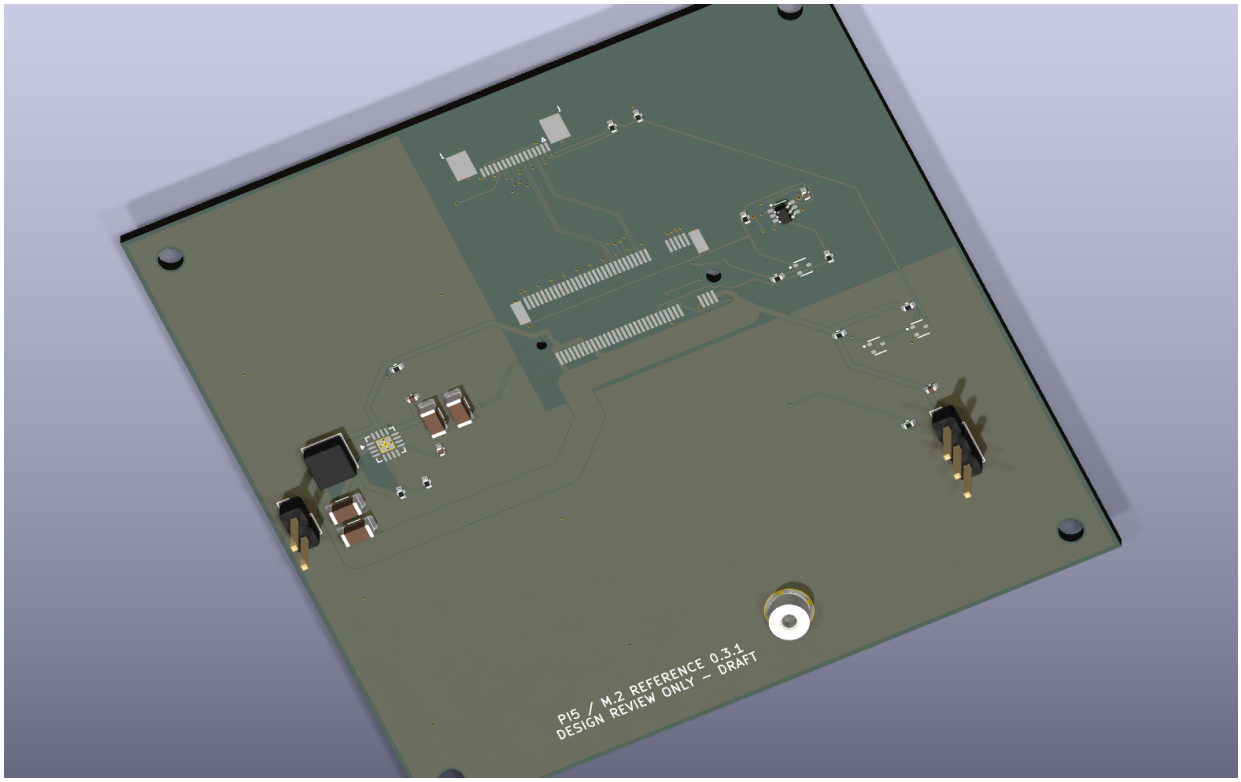
Thermal model is a 2D sheet with assumed heat/convection and simplified package/via path. No measured Tj, SSD die temperature, enclosure airflow or full 3D validation. Current detailed signed tables: hardware/reference_pcie_adapter/INDEPENDENT_REVIEW.md.

Process review and release limits

Publish truthful engineering evidence; retain hardware holds

Stack, width/gap, copper, dielectric and solder-mask sensitivity were rerun. The manufacturer reference uses 40.64 um copper, 30.48/15.24 um mask with Er3.8 and 17.78 um top narrowing; its stack table separately lists 35 um copper. Inputs are not a manufacturing commitment.

A 7628 candidate with h0.2104 mm, Er4.4, width0.30/gap0.20 mm passes the original two corners at 81.830-97.575 ohm. Adding published copper/mask/etch geometry produces a 79.257 ohm low corner: -1.743 ohm margin. It was not adopted. No requirement or tolerance was relaxed.



Open hardware gates: joint voltage envelope, effective capacitors, thermal exit/SSD limits, qualified stack/process, connector/cable/assembly, full 3D channel and fabrication review.

Ready: documented RTL/model source, 63-test regression, 33 E2E scenarios, generic synthesis, debug evidence and portable plots.
Not tested: FPGA timing/implementation, physical DDR/PCIe, measured throughput/power, hardware bring-up or fabrication. Historical v4 slides/PDFs are excluded from the current release scope.

Official references: TI tps62130.pdf and spra953d.pdf; JLC stack table and impedance-calculator guide. Exact links, assumptions and evidence are in the independent PCB review. Reproduce: docs/REPRODUCING.md. Final disposition: docs/REQUIREMENTS_MATRIX.md. No purchases, fabrication or GitHub push.